

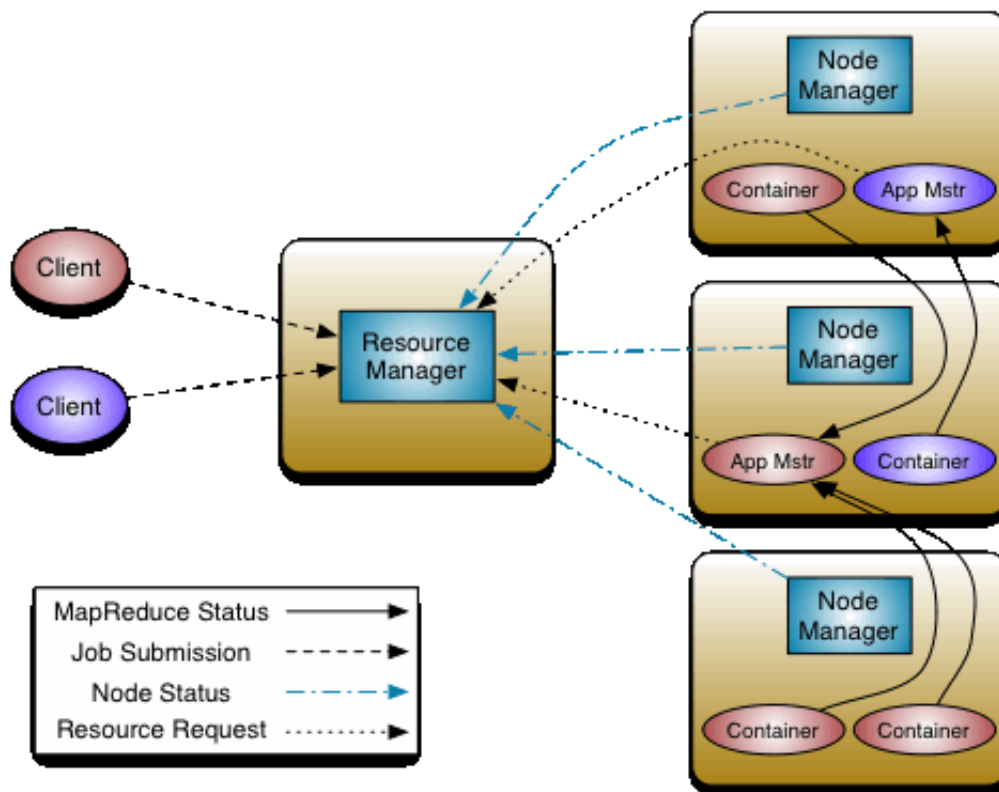
Apache Hadoop NextGen MapReduce (YARN)

MapReduce has undergone a complete overhaul in hadoop-0.23 and we now have, what we call, MapReduce 2.0 (MRv2) or YARN.

The fundamental idea of MRv2 is to split up the two major functionalities of the JobTracker, resource management and job scheduling/monitoring, into separate daemons. The idea is to have a global Resource Manager (*RM*) and per-application ApplicationMaster (*AM*). An application is either a single job in the classical sense of Map-Reduce jobs or a DAG of jobs.

The Resource Manager and per-node slave, the Node Manager (*NM*), form the data-computation framework. The Resource Manager is the ultimate authority that arbitrates resources among all the applications in the system.

The per-application ApplicationMaster is, in effect, a framework specific library and is tasked with negotiating resources from the Resource Manager and working with the Node Manager(s) to execute and monitor the tasks.



The Resource Manager has two main components: Scheduler and ApplicationsManager.

The Scheduler is responsible for allocating resources to the various running applications subject to familiar constraints of capacities, queues etc. The Scheduler is pure scheduler in the sense that it performs no monitoring or tracking of status for the application. Also, it offers no guarantees about restarting failed tasks either due to application failure or hardware failures. The Scheduler performs its scheduling function based on the resource requirements of the applications; it does so based on the abstract notion of a resource *Container* which incorporates elements such as memory, cpu, disk, network etc. In the first version, only memory is supported.

The Scheduler has a pluggable policy plug-in, which is responsible for partitioning the cluster resources among the various queues, applications etc. The current Map-Reduce schedulers such as the CapacityScheduler and the FairScheduler would be some examples of the plug-in.

The CapacityScheduler supports hierarchical queues to allow for more predictable sharing of cluster resources

The ApplicationsManager is responsible for accepting job-submissions, negotiating the first container for executing the application specific ApplicationMaster and provides the service for restarting the ApplicationMaster container on failure.

The NodeManager is the per-machine framework agent who is responsible for containers, monitoring their resource usage (cpu, memory, disk, network) and reporting the same to the Resource Manager/Scheduler.

The per-application ApplicationMaster has the responsibility of negotiating appropriate resource containers from the Scheduler, tracking their status and monitoring for progress.

MRV2 maintains **API compatibility** with previous stable release (hadoop-1.x). This means that all Map-Reduce jobs should still run unchanged on top of MRv2 with just a recompile.